



Chemistry 101: Stoichiometry in the Lab — A Full Limiting-Reactant Session

Chemistry 101 • Chemistry

COURSE-OBJ.CHEM101.2

Balance equations, identify a limiting reactant from given masses, and calculate theoretical and percent yield, applying each step to real lab data. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

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Plan the session

Course and objective

Chemistry 101 · Chemistry; COURSE-OBJ.CHEM101.2 — Balance chemical equations, identify the limiting reactant given starting masses, and calculate theoretical and percent yield from lab data.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both lab notebook passages + vocabulary 15 min · guided limiting-reactant organizer 18 min · partner yield-calculation discussion 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.



Before you begin

Key terms

balanced equation — an equation with equal atoms of each element on both sides

mole ratio — the ratio of moles of substances in a balanced equation

limiting reactant — the reactant that runs out first and caps product formed

excess reactant — the reactant left over after the reaction completes

theoretical yield — the maximum possible product mass from the limiting reactant

percent yield — actual yield divided by theoretical yield, times 100

Concept focus

A balanced equation shows the mole ratio in which reactants combine and products form; coefficients must be adjusted so the same number of each atom appears on both sides. Given starting masses, converting each reactant to moles and comparing to the balanced ratio identifies the limiting reactant — the one that runs out first and caps how much product can form; any reactant left over is in excess. Theoretical yield is the maximum product mass calculable from the limiting reactant; percent yield compares the mass actually collected in the lab to that theoretical maximum, and is always reported below 100% due to real losses (transfer, side reactions, purification).

Model 1 — Balancing an equation

Unbalanced: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$. Balanced: $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$. Check: 4 Fe atoms and 6 O atoms appear on both sides.

Model 2 — Finding the limiting reactant

Given 2.0 mol H_2 and 2.0 mol O_2 for $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (a 2:1 ratio needed): 2.0 mol H_2 needs only 1.0 mol O_2 , so O_2 is in excess and H_2 is the limiting reactant.

WARM-UP

Warm-up

Answer using what you already know or recall from the previous session.

1. Balance this equation and show your reasoning: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$.

2. In the lab's balanced equation $\text{CuSO}_4 + 2\text{NaOH} \rightarrow \text{Cu}(\text{OH})_2 + \text{Na}_2\text{SO}_4$, what mole ratio of NaOH to CuSO_4 is required for a perfect, no-leftover reaction?



Lab Notebook, Part 1: Balancing and Planning the Reaction

Today's lab produces copper(II) hydroxide by reacting copper(II) sulfate with sodium hydroxide in solution: $\text{CuSO}_4 + \text{NaOH} \rightarrow \text{Cu(OH)}_2 + \text{Na}_2\text{SO}_4$. Before touching any chemicals, every student must balance this equation and use it to plan exactly how much of each reactant the lab requires.

Working through it: one copper, one sulfate, so those stay balanced with a coefficient of 1 on CuSO_4 and Na_2SO_4 . But sodium and hydroxide each need a coefficient of 2 to balance the two sodium atoms and two hydroxide groups on the product side: $\text{CuSO}_4 + 2\text{NaOH} \rightarrow \text{Cu(OH)}_2 + \text{Na}_2\text{SO}_4$. Checking every atom: 1 Cu, 1 S, 4 O (from sulfate) + 2 O (from hydroxide) = 6 O total on the left; on the right, 1 Cu, 2 O + 2 H (from Cu(OH)_2), plus 2 Na, 1 S, 4 O (from sulfate) — all atoms match once every O is counted carefully.

Today's planned amounts: 0.0500 mol CuSO_4 and 0.0800 mol NaOH . The balanced equation requires NaOH at twice the moles of CuSO_4 (a 1:2 ratio) for a perfect, no-leftover reaction. Students are asked to determine, before mixing anything, which reactant will run out first at these specific starting amounts.



Lab Notebook, Part 2: Measuring Yield

After calculating that CuSO_4 was the limiting reactant (0.0500 mol CuSO_4 would need 0.1000 mol NaOH to fully react, but only 0.0800 mol NaOH was available — wait, students are asked to check this calculation carefully against the ratio, since it determines everything that follows), the lab groups proceed to run the reaction, filtering out the solid copper(II) hydroxide precipitate that forms.

Using the limiting reactant's moles and the 1:1 mole ratio between CuSO_4 and $\text{Cu}(\text{OH})_2$ in the balanced equation, each group calculates a theoretical yield: the maximum possible mass of $\text{Cu}(\text{OH})_2$ the reaction could produce if every bit of the limiting reactant converted perfectly, with nothing lost. $\text{Cu}(\text{OH})_2$ has a molar mass of 97.57 g/mol.

After filtering, drying, and weighing their actual product, Group 3 collects 3.62 grams of $\text{Cu}(\text{OH})_2$. Group 7, running the identical planned reaction, collects only 3.05 grams. Both groups are asked to calculate their percent yield and explain, using specific possible lab errors, why two groups running the same planned reaction might collect different actual masses even though their theoretical yield calculation should be identical.



Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, calculate the theoretical yield in grams using the molar mass given (97.57 g/mol), showing your unit conversion.

2. With a partner, calculate Group 3's percent yield (actual 3.62 g) and Group 7's percent yield (actual 3.05 g), showing the percent-yield formula used.

3. Discuss: why does the lesson say percent yield is "always reported below 100%"? Is a percent yield over 100% possible, and what would it suggest if reported?



Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Propose two specific, different lab errors that could explain why Group 7's yield (78.2%) was lower than Group 3's (92.8%), given they ran the identical planned reaction.

2. Explain, using the definition of excess reactant, what happens to the leftover NaOH or CuSO₄ (whichever is not limiting) in this reaction.

3. Calculate how many moles of the excess reactant remain unreacted after the reaction goes to completion.



Independent practice

Work independently. Show your work or cite evidence in every answer.

4. If a student mistakenly used CuSO_4 (the excess reactant) instead of NaOH to calculate theoretical yield, would their theoretical yield be too high or too low? Explain your reasoning.

5. Explain why chemists check that every atom balances on both sides of an equation, using a specific atom count from the copper hydroxide equation as evidence.

6. Design an original limiting-reactant scenario (different reactants, different starting moles) and identify which reactant is limiting, showing your ratio comparison.



Small-group extension work

Use for review or additional application practice.

1. Using your scenario from the previous question, calculate the theoretical yield of product in moles.

2. Write a two-to-three sentence lab-report explanation (in your own words) of why two groups running the "same" planned reaction can reasonably report different percent yields, without implying either group made a scientific error.



Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Compare theoretical yield and percent yield: what does each one tell you, and why do chemists need both numbers rather than just one?

2. Explain why molar mass (97.57 g/mol for $\text{Cu}(\text{OH})_2$) is needed as a conversion factor, and what two units it connects.



Assessment

Complete independently to show what you learned today.

1. Self-assessment: rate your confidence in identifying a limiting reactant from starting moles, and name one specific step in the process you want to practice again.



Assessment · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.



Answer key

1. $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$. Reasoning: balancing Fe_2O_3 requires an even number of Fe, so start with $2\text{Fe}_2\text{O}_3$, giving 4 Fe and 6 O on the right, which requires 4Fe and 3O_2 on the left.
2. 2:1 (NaOH to CuSO_4) — every 1 mole of CuSO_4 requires 2 moles of NaOH.
3. $0.0500 \text{ mol CuSO}_4 \times 2 = 0.1000 \text{ mol NaOH}$ needed; only 0.0800 mol NaOH is available, which is less than needed.
4. NaOH is the limiting reactant, because the reaction needs 0.1000 mol of it to fully react with the available CuSO_4 , but only 0.0800 mol is present — NaOH will run out first.
5. This is the same error as jumping to a conclusion without doing the mole-ratio comparison; the actual calculation shows NaOH, not CuSO_4 , runs out first, which reverses which reactant's moles must be used for every later yield calculation — getting this step wrong invalidates the rest of the analysis.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. The limiting reactant is NaOH at 0.0800 mol; the ratio of NaOH to Cu(OH)₂ is 2:1, so $0.0800 \text{ mol NaOH} \div 2 = 0.0400 \text{ mol Cu(OH)}_2$ theoretically produced.
7. $0.0400 \text{ mol Cu(OH)}_2 \times 97.57 \text{ g/mol} = 3.9028 \text{ g}$, or about 3.90 g theoretical yield.
8. Percent yield = (actual $\div$ theoretical) $\times$ 100. Group 3: $(3.62 \div 3.90) \times 100 \approx 92.8\%$. Group 7: $(3.05 \div 3.90) \times 100 \approx 78.2\%$.
9. Real reactions lose some product to transfer, side reactions, or incomplete purification, so 100% recovery is essentially never achieved; a reported yield over 100% would suggest a measurement or purity error (e.g., the product still contains water or leftover reactant adding extra mass), not a genuinely more-than-complete reaction.
10. Answers vary; must propose two specific, plausible causes, e.g., some precipitate lost during filtering (stuck to filter paper or lost in transfer), incomplete drying leaving residual water weighed as part of the "dry" product (which would raise, not lower, apparent yield — so likely product loss instead), or a measurement error in reactant amounts before mixing.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. The excess reactant does not fully react; some of it remains unreacted in the mixture after the limiting reactant is used up, since there isn't enough of the limiting reactant to consume all of the excess reactant.

12. CuSO_4 needed to react with $0.0800 \text{ mol NaOH} = 0.0800 \div 2 = 0.0400 \text{ mol CuSO}_4$ consumed; starting CuSO_4 was 0.0500 mol , so $0.0500 - 0.0400 = 0.0100 \text{ mol CuSO}_4$ remains unreacted (excess).

13. Too high — using CuSO_4 's full 0.0500 mol (instead of the smaller, correctly limiting NaOH -derived 0.0400 mol) would produce a larger, incorrect theoretical yield, since it assumes a reactant is available in an amount it is not able to actually be converted to product given the true limiting reactant.

14. The law of conservation of mass requires that atoms are neither created nor destroyed in a chemical reaction, only rearranged; checking, for example, that 6 oxygen atoms appear on both sides of $\text{CuSO}_4 + 2\text{NaOH} \rightarrow \text{Cu(OH)}_2 + \text{Na}_2\text{SO}_4$ confirms the equation obeys this law and accurately represents what actually happens.

15. Answers vary; must state a balanced equation's mole ratio, give specific starting moles for two reactants, and correctly identify the limiting reactant using a ratio comparison consistent with the given numbers.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. Answers vary; must correctly apply the mole ratio between the limiting reactant and the product from the student's own balanced equation.

17. Answers vary; must reflect that some product loss during transfer/filtering/drying is a normal, expected part of real lab technique and does not by itself indicate the reaction or the underlying chemistry was done incorrectly.

18. Theoretical yield tells you the maximum possible product mass a reaction could produce under ideal conditions; percent yield tells you how efficiently the actual lab procedure captured that maximum. Both are needed because percent yield alone (without a theoretical yield benchmark) would have no basis for comparison, and theoretical yield alone does not reflect real lab performance.

19. Molar mass connects grams and moles; since lab balances measure mass but the balanced-equation mole ratio works in moles, molar mass is the conversion factor needed to move between the two units in either direction.

20. Answers vary; must name a specific step from the process (e.g., converting to moles, applying the mole ratio, comparing the two reactants) to revisit.

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment · accessed September 18, 2026

COURSE-OBJ.CHEM101.2 — Balance chemical equations, identify the limiting reactant given starting masses, and calculate theoretical and percent yield from lab data. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

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